

A 1770- μm^2 Leakage-Based Digital Temperature Sensor With Supply Sensitivity Suppression in 55-nm CMOS

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Abstract—This article presents a leakage-based digital temperature sensor with reduced supply sensitivity for on-chip thermal management. The sensor, featured with a novel supply sensitivity suppression mechanism, performs the temperature-to-frequency conversion by a leakage-dominated ring oscillator (LDRO) with exponential temperature dependence. Thanks to the proposed robust and reconfigurable Schmitt-trigger-based delay cell, both NMOS and PMOS leakage-based sensors can be evaluated in a single design. Fabricated in a standard 55-nm CMOS digital process, the proposed digital temperature sensor occupies a silicon area of only 1770 μm^2 and can operate under a supply ranging from 0.8 to 1.3 V, with the supply sensitivities of 2.53–5.22 $^{\circ}\text{C}/\text{V}$ and 2.84–5.76 $^{\circ}\text{C}/\text{V}$ in two working modes at room temperature, respectively. Measurement results show that the sensor achieves an inaccuracy of ± 0.70 $^{\circ}\text{C}$ (3σ) from -40 $^{\circ}\text{C}$ to 125 $^{\circ}\text{C}$ after two-point calibration.

Index Terms—CMOS temperature sensor, leakage, ring oscillator (RO), supply sensitivity, thermal management.

I. INTRODUCTION

CMOS temperature sensors are increasingly important for thermal management of modern high-performance microprocessors and systems-on-chips (SoCs), in which tens of sensors are implemented on the same chip for real-time cold and hot spots detection [1], [2]. Due to the requirement of low-cost multi-site integration, temperature sensors with small area (< 0.01 mm^2) and low power (microwatt-level) are highly desirable [3], [4]. Furthermore, it is essential for temperature sensors to operate under a local (digital) supply voltage in order to reduce the overhead associated with the routing of a separate power supply voltage [5]. However, due to the application of dynamic voltage scaling (DVS) in modern microprocessors, the digital supply may have a dc variation of hundreds of millivolts. Also, considering the noisy digital circuits, sensors directly connected to the digital supply

suffer from tens of millivolt ripples in the supply voltage. In this scenario, temperature sensors should also have enough supply sensitivity rejection to meet the moderate accuracy requirement (e.g., less than ± 3 $^{\circ}\text{C}$) under a wide supply range.

Various types of CMOS temperature sensors have been studied [6]. bipolar junction transistor (BJT)-based sensors are used widely due to their high accuracy of a few tenths of a degree over a wide temperature range from -55 $^{\circ}\text{C}$ to 125 $^{\circ}\text{C}$ [7], [8]. However, they are incompatible with the sub-1-V digital supply in advanced processes. Popular alternatives to BJT sensors are resistor-based designs with high energy efficiency and high resolution, but passive devices, such as resistors and capacitors, become the bottleneck of further area reduction [9], [10]. Although thermal diffusivity (TD) sensors with small areas can operate under a sub-1-V digital supply voltage, their milliwatt-level power dissipation will cause self-heating, which introduces additional inaccuracy [11], [12].

Recently, time-domain all-MOSFET temperature sensors, which employ the temperature dependence of delay lines, are also reported [5], [13]–[17]. Among them, ring-oscillator (RO)-based digital temperature sensors that can operate with a sub-1-V supply voltage, sub-0.01- mm^2 area, and microwatt-level power consumption are candidates for thermal management [5], [13]–[15]. However, it is still challenging for the RO itself to be tolerant to supply voltage variations [16]. Although high-frequency ac variations in the supply voltage can be integrated and averaged when $f_{\text{noise}} > f_{\text{RO}}$, dc variations still influence the RO seriously. To solve this issue, several techniques have been proposed in previous work. In general, they can be classified into two categories, as shown in Fig. 1(a) and (b):

- 1) generating a virtual supply voltage (V_{VDD}) with a local voltage regulator [13]–[15];
- 2) evaluating the ratio of the output frequencies of two different ROs, which have similar supply sensitivities [5].

However, local voltage regulators require extra analog operational amplifiers and bandgap reference current sources [13], [14], which increase the design cost for multi-site integration in SoCs. Stacking native IO devices as a simple voltage regulator can be alternatively used in [15], yet such a device requires additional design cost and may not be always available in every CMOS process. On the other hand, evaluating the ratio of the output frequencies of two

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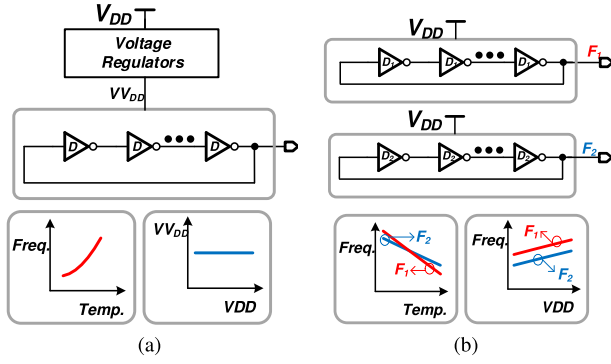


Fig. 1. (a) Diagram of prior art 1 in [13]–[15] (b) Diagram of prior art 2 in [5].

ROs with similar supply sensitivities relies on the matching of two different ROs, resulting in a limited supply sensitivity of 34 °C/V [5].

To solve this issue, a new digital temperature sensor based on a leakage-dominated RO (LDRO) with supply sensitivity suppression is presented. Due to the embedded supply sensitivity rejection mechanism, the sensor can operate directly under a supply voltage from 0.8 to 1.3 V without regulators. In addition, a robust and reconfigurable Schmitt-trigger-based delay cell is proposed in this article, and thus, both NMOS and PMOS leakage-based sensors can be evaluated in a single design. A prototype sensor, implemented in a standard 55-nm CMOS digital process, occupies a silicon area of only 1770 μm^2 . It achieves a supply sensitivity of 2.53–5.22 °C/V/2.84–5.76 °C/V and an accuracy of ± 0.70 °C/ ± 0.94 °C (3σ) from -40 °C to 125 °C in two working modes.

This article is organized as follows. Section II introduces the basic operating principle of the proposed digital temperature sensors. Section III describes the circuitry implementation details, followed by the measurement results in Section IV. Section V concludes this article.

II. OPERATING PRINCIPLE

The supply sensitivity of an RO-based temperature sensor (°C/V) can be calculated by ratio of RO's supply sensitivity ($1/F_{RO} \cdot dF_{RO}/dV$, %/V) to its temperature sensitivity ($1/F_{RO} \cdot dF_{RO}/dT$, %/°C). An RO with a low supply sensitivity and a high temperature sensitivity can perform direct temperature-to-frequency conversion under a digital supply voltage with enough supply sensitivity rejection, thus reducing the complexity of the sensor from the system level. Furthermore, it is highly desired that the temperature dependence of the RO's frequency can be linearized easily, so as to reduce the calibration cost. To satisfy these conditions, an all-MOSFET-based LDRO with exponential temperature dependence and suppressed supply sensitivity is proposed.

A. Leakage-to-Frequency Conversion

With the continuous scaling down of CMOS technologies, the ever-increasing leakage current in MOS devices is undesirable and challenging for advanced circuit design [18], [19].

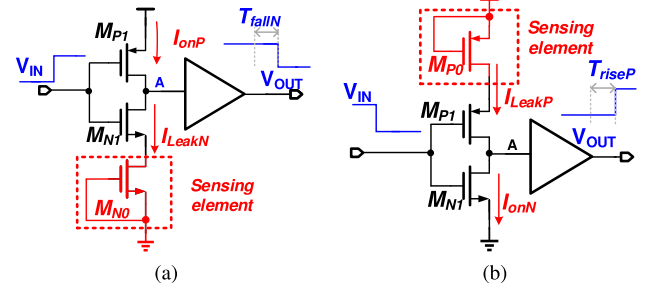


Fig. 2. (a) Diagram of the proposed leakage-based N-mode delay cell. (b) Diagram of the proposed leakage-based P-mode delay cell.

However, the exponential temperature dependence of the leakage current is an excellent characteristic for temperature sensing [20]. The subthreshold leakage current is a significant portion of total leakage and depends exponentially on temperature, which is a viable solution for temperature sensing in nanometer CMOS technologies [20], [21]. If an NMOS transistor's gate and source are both connected to the ground (GND), the subthreshold leakage current I_{LeakN} can be modeled as [22]

$$I_{LeakN} = I_0 \cdot \exp\left(\frac{-V_{th}}{nV_T}\right) \cdot \left(1 - \exp\left(\frac{-V_{ds}}{V_T}\right)\right) \quad (1)$$

where $I_0 = \mu C_{ox}(W/L)V_T^2 e^{1.8}$, μ is the NMOS mobility, C_{ox} is the oxide capacitance, V_{th} is the threshold voltage of the transistor, n is the subthreshold swing coefficient, V_T is the thermal voltage and equals to kT/q , k is Boltzmann's constant, q is the electron charge, T is the temperature in kelvin, and V_{ds} is the drain–source voltage drop.

The subthreshold leakage current can be converted to a delay by charging or discharging a capacitive load [20]. Fig. 2(a) shows the simplified NMOS leakage-based delay cell (NLDC) in this article. A normal inverter is in series with a “disabled” NMOS footer, followed by a digital buffer, which can be simply implemented by two cascaded inverters or a Schmitt trigger. Similar to the working principle in [21], when V_{IN} goes from low to high, node A can only be pulled low by the leakage current of M_{N0} slowly, and the output of the buffer V_{OUT} will keep logic high until V_A is smaller than the threshold voltage of the buffer, thus generating an NMOS-leakage-dominated fall delay t_{fallN} . If V_{IN} changes from high to low, node A will be pulled high quickly due to the strong pull-up of transistor M_{P1} , and thus, the rise delay t_{riseN} is negligible when compared with t_{fallN} . A PMOS leakage-based delay cell (PLDC) also works similarly, as shown in Fig. 2(b).

The detailed temperature sensitivity of t_{fallN} can be obtained by an equivalent circuit analysis with three phases: reset phase, quick charging phase, and leakage-dominated discharging phase.

The reset phase sets the initial state for the delay cell. Fig. 3(a) shows the equivalent circuit of the NLDC in the reset phase. The “disabled” NMOS M_{N0} is modeled as a resistor R_O with adjustable resistance. M_{N1} and M_{P1} are modeled as two switches SW_N and SW_P . C_1 and C_2 are the parasitic capacitances at nodes B and A, respectively. When

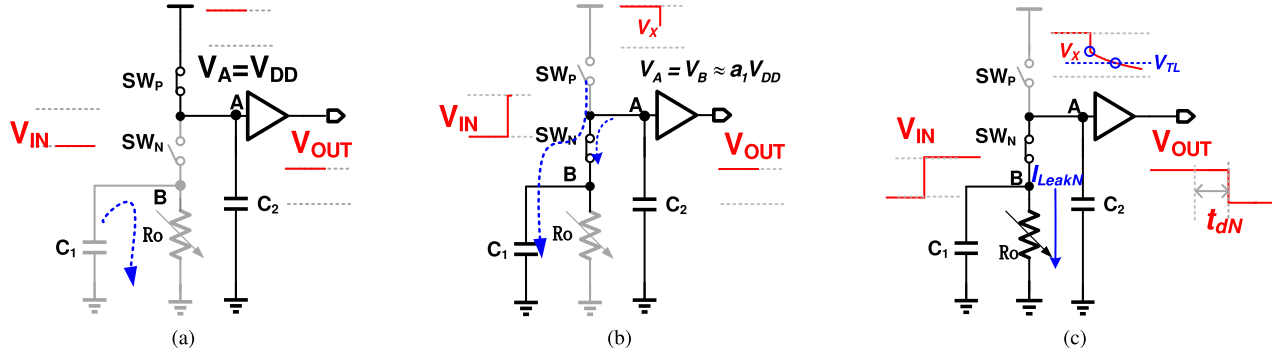


Fig. 3. (a) Equivalent circuit in the reset phase for the N-mode delay cell. (b) Equivalent circuit in the quick charging phase for the N-mode delay cell. (c) Equivalent circuit in the leakage-dominated discharging phase for the N-mode delay cell.

V_{IN} is logic low, SW_P is turned on and SW_N is turned off. During this phase, the voltage at node B is pulled down slowly by the leakage current of the “disabled” NMOS M_{N0} and the voltage at node A is reset to V_{DD} , and thus, the delay cell has an output of logic high.

When V_{IN} changes from logic low to logic high, SW_P is turned off and SW_N is turned on. During the switching of SW_P and SW_N , the short-circuit current through the switches charges C_1 at node B, causing the voltage at node B to rise quickly. Next, the charge in C_1 and C_2 redistributes quickly until the voltages at nodes A and B achieve balance. Since the charge caused by the short-circuit current during switching as well as the charge coming from C_1 and C_2 in the reset phase are supply voltage-dependent, the balanced voltage at nodes A and B can be simplified as $V_X \approx a_1 \cdot V_{DD}$ to the first-order approximation, where a_1 is the supply sensitivity of V_X . As shown in Fig. 3(b), the quick charging phase takes a negligible time and generates a voltage change at node A.

If V_X is higher than the high-to-low threshold voltage V_{TL} of the buffer, the total capacitor $C_L = C_1 + C_2$ is discharged by the leakage current of the “disabled” NMOS M_{N0} slowly until the voltage at node A reaches V_{TL} . Fig. 3(c) shows the diagram of the leakage-dominated discharging phase. The differential equation during this phase is given by

$$C_L \frac{dV_A}{dt} = -I_{leakN}. \quad (2)$$

If V_{TL} is larger than $3V_T$, (1) can be rewritten as

$$I_{leakN} = I_0 \cdot \exp\left(\frac{-V_{th}}{nV_T}\right). \quad (3)$$

According to (2) and (3), the leakage-dominated discharging time t_{dN} is as follows:

$$t_{dN} = C_L \cdot \frac{V_X - V_{TL}}{I_0 \cdot \exp\left(\frac{-V_{th}}{nV_T}\right)} \quad (4)$$

where $V_T = kT/q$. Equation (4) reveals that t_{dN} has an exponential temperature dependence and can be rewritten as

$$\ln(t_{dN}) = K + \frac{V_{th} \cdot q}{nkT} \quad (5)$$

where $K = \ln(C_L \cdot ((V_X - V_{TL})/I_0))$. To be noted, although I_0 is temperature-dependent, the temperature dependence of I_0

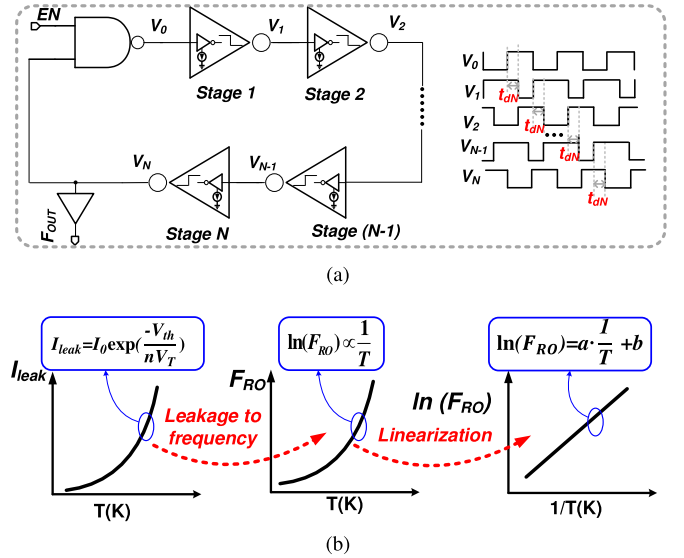


Fig. 4. (a) Diagram of the proposed n-type LDRO. (b) Operating principle of the n-type LDRO.

will be suppressed by the logarithmic operation. Thus, $\ln(t_{dN})$ is approximately proportional to $1/T$. In addition, since the quick charging phase takes a negligible time, the total fall delay t_{fallN} is approximately equal to t_{dN} .

As shown in Fig. 4(a), an n-type LDRO is generated using an even number of NLDCs and a NAND gate. Because the delay in one NLDC is only dominated by the leakage-dominated discharging time t_{dN} , the oscillating frequency is approximately expressed as follows:

$$F_{RO} = \frac{1}{N \cdot t_{riseN} + N \cdot t_{fallN}} \approx \frac{1}{N \cdot t_{dN}} \quad (6)$$

where N is the number of NLDCs. In this architecture, the local mismatch in each stage can be averaged. In addition, $\ln(F_{RO})$ is also linear with $1/T$ according to (5) and (6). Thus, the exponentially temperature-dependent frequency can be linearized as follows, which is also applied in [15]:

$$\ln(F_{RO}) = a \cdot \frac{1}{T} + b \quad (7)$$

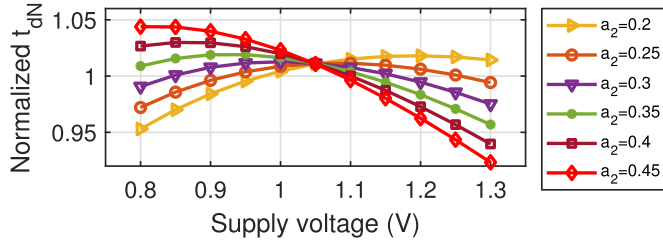


Fig. 5. Mathematical model simulation of the normalized delay t_{dN} by varying a_2 when $\lambda = 0.1$, $n = 1.5$, $V_T = 26$ mV, and $a_1 = 0.5$.

where a and b are the fitting coefficients for the frequency to temperature conversion. Even though the exponentially temperature-dependent frequency varies due to process variations, (7) reveals that a two-point calibration can be applied to calculate a and b for each die. Fig. 4(b) summarizes the basic operating principle of the proposed LDRO. Similarly, a p-type LDRO, which is based on PLDCs, can also convert the leakage to a frequency with exponential temperature dependence. After the leakage-to-frequency conversion, a frequency-to-digital converter (FDC) can be applied in the digital domain.

B. Analysis of the Supply Sensitivity

The supply sensitivity suppression mechanism of the temperature sensor is analyzed in this article. Take n-type LDRO for example, since the subthreshold leakage current I_{leakN} increases with V_{DD} due to the drain-induced barrier lowering (DIBL) effect [23], and the voltage swing ($V_X - V_{TL}$) for charging can also be designed to have positive supply sensitivity and it is possible to cancel the supply sensitivities of I_{leakN} and $V_X - V_{TL}$, generating a supply independent delay. The detailed analysis is as follows.

Taking DIBL effect into consideration, (3) is rewritten as

$$I_{\text{leakN}} = I_0 \cdot \exp\left(\frac{-V_{th0} + \lambda V_{ds}}{nV_T}\right) \quad (8)$$

where λ is the DIBL coefficient and is around 0.1 in the target 55-nm CMOS process. Equation (2) is resolved as follows:

$$-\frac{C_L}{I_0} \cdot \exp\left(\frac{V_{th0}}{nV_T}\right) \cdot \int_{V_X}^{V_{TL}} \exp\left(\frac{-\lambda V_A}{nV_T}\right) dV_A = \int_0^{t_{dN}} dt \quad (9)$$

$$t_{dN} = K_1 \cdot \left[\exp\left(-\frac{\lambda V_{TL}}{nV_T}\right) - \exp\left(-\frac{\lambda V_X}{nV_T}\right) \right] \quad (10)$$

where $K_1 = (C_L \cdot nV_T) / \lambda I_0 \cdot \exp(V_{th0}/nV_T)$ and $V_X = a_1 V_{DD}$. If the threshold voltage V_{TL} of the buffer is also proportional to V_{DD} and can be expressed as $V_{TL} = a_2 \cdot V_{DD}$, where a_2 is the supply sensitivity of V_{TL} and $a_2 < a_1$ to ensure that $V_{TL} < V_X$, (11) can be rewritten as a function of a_1 and a_2 and V_{DD}

$$t_{dN} = K_1 \cdot \left[\exp\left(-\frac{\lambda \cdot a_2 \cdot V_{DD}}{nV_T}\right) - \exp\left(-\frac{\lambda \cdot a_1 \cdot V_{DD}}{nV_T}\right) \right]. \quad (11)$$

Clearly, the supply sensitivity can be optimized by setting proper coefficients a_1 and a_2 . For example, fixing a_1 to 0.5 and varying a_2 from 0.2 to 0.45, the supply sensitivity of t_{dN} is

simulated according to (11) over a supply voltage ranging from 0.8 to 1.3 V, as shown in Fig. 5. It reveals that as a_2 increases, the normalized delay t_{dN} changes from increasing with V_{DD} to decreasing with V_{DD} . When a_2 is 0.3, the supply sensitivity is reduced to 7.5%/V over a supply voltage ranging from 0.8 to 1.3 V. Using finer steps and further optimizing a_1 , smaller supply sensitivity can be achieved theoretically.

In order to estimate the supply sensitivity of the whole sensor, both the temperature and the supply sensitivity of the LDRO are modeled according to the previous mathematical analysis. As shown in (4), coefficients a_1 and a_2 affect $V_X - V_{TL}$. However, they have no impact on the normalized temperature sensitivity ($1/F_{RO} \cdot dF_{RO}/dT$, %/°C) at a given temperature point, which can be derived from (4) and (6) as follows:

$$\frac{1}{F_{RO}} \cdot \frac{dF_{RO}}{dT} = \frac{V_{th} \cdot q}{nkT^2}. \quad (12)$$

For simplicity, the temperature sensitivity at room temperature that is around 4.6%/°C according to the mathematical model simulations is used to convert the supply sensitivity in frequency to voltage ($1/F_{RO} \cdot dF_{RO}/dV$, %/V) to that in temperature to voltage (°C/V). The supply sensitivity of the n-type LDRO is modeled by varying a_1 and a_2 under a supply voltage ranging from 0.8 to 1.3 V. The mathematical model simulation results are shown in Fig. 6. It shows that for any $a_1 \in (0.4, 0.8)$, the supply sensitivity can always be reduced to less than 2 °C/V by optimizing a_2 . In addition, given process variations and component mismatches in circuit implementations, which may cause diversions of a_1 and a_2 , the proposed supply sensitivity suppression technique can reduce the supply sensitivity to an acceptable range theoretically. For example, considering $a_1 = 0.5$ and $a_2 = 0.28$ with $\pm 10\%$ variations, it shows a maximum supply sensitivity of 2.6 °C/V under a supply voltage with 500-mV dc variation. Similarly, the proposed supply sensitivity suppression technique can be applied to a p-type LDRO too.

III. CIRCUIT IMPLEMENTATION

A. Overall Architecture

Based on the analysis, a prototype digital temperature sensor is implemented in a standard 55-nm CMOS process. Fig. 7 shows the system diagram of the proposed digital temperature sensor. A reconfigurable LDRO consisting of four leakage-based delay cells along with a NAND gate operates as a sensing core. Due to the digital-friendly LDRO, the sensing frequency can be processed by the digital part directly. Thus, the LDROs can be used as “standard cells” and be placed near potential hot spots. To experiment with the feasibility of multi-site integration in an SoC, eight LDROs share the same FDC through a multiplexer. Furthermore, to further suppress the systematic error caused by residual non-linearity, a polynomial fitting with fixed coefficients can be applied to remove the systematic error [5], [9], which can be performed by fitting the measured temperature with the reference temperature using several chips during the testing. Then, the coefficients can be fixed for the entire lot. Both the logarithmic frequency-to-temperature conversion and the systematic error correction

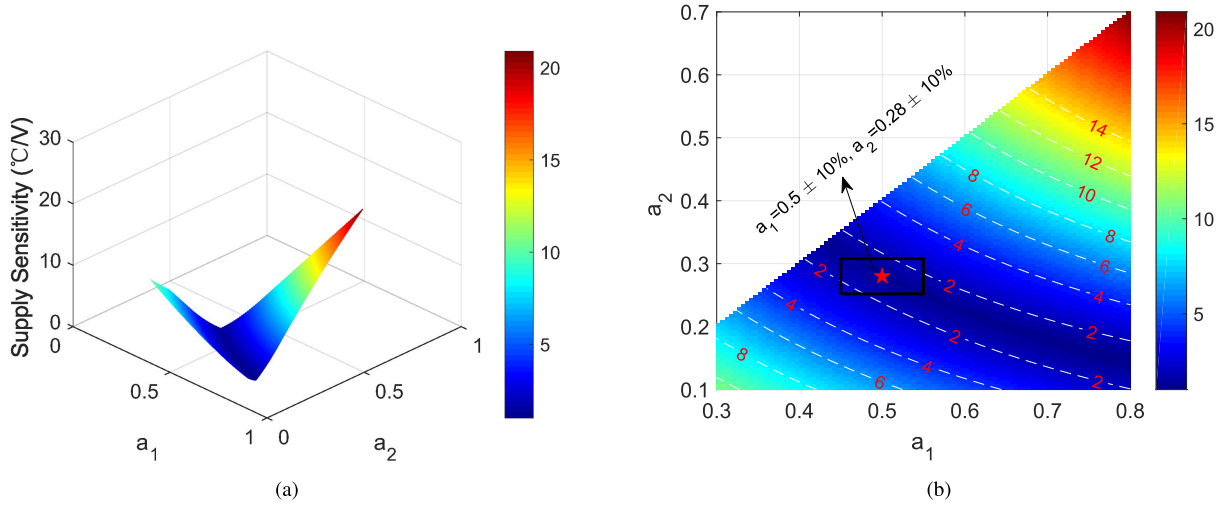


Fig. 6. (a) Mathematical model simulation of the supply sensitivity by varying a_1 and a_2 over a supply voltage ranging from 0.8 to 1.3 V when $\lambda = 0.1$, $n = 1.5$, $V_T = 26$ mV, and $V_{th0} = 500$ mV. (b) Vertical view with contour lines.

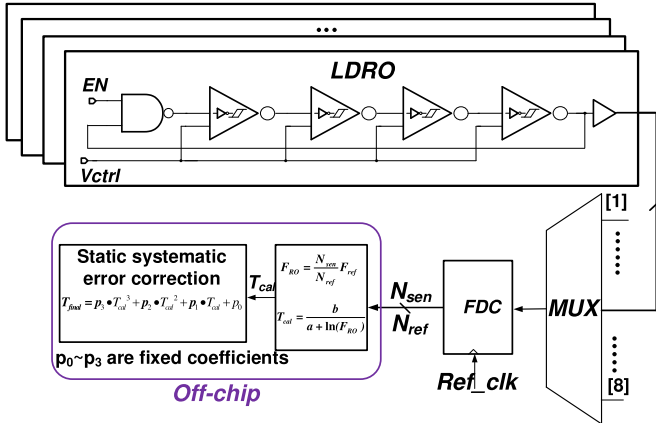


Fig. 7. System diagram of the proposed digital temperature sensor.

are performed off-chip for flexibility in this design. They can also be done by the microprocessor without further hardware cost.

B. Reconfigurable LDRO

Fig. 8 shows the circuit diagram of the reconfigurable Schmitt-trigger-based delay cell in this article, which consists of a power-gated inverter and a Schmitt trigger. A typical inverter is in series with a PMOS header M_{P0} and an NMOS footer M_{N0} in the delay cell, which can be configured into N-mode or P-mode by a 1-bit control signal. In the N-mode, M_{P0} is enabled and M_{N0} is disabled. The voltage at node A is then compared with the high-to-low threshold voltage V_{TL} of the Schmitt trigger, which is equivalent to that in Fig. 2(a). Similarly, when M_{P0} is disabled and M_{N0} is enabled, the low-to-high threshold voltage V_{TH} is used and the delay cell operates in the P-mode. Because of the hysteresis of the Schmitt trigger, both the NMOS-based and PMOS-based sensors can be evaluated in a single design. As shown in Fig. 8, another benefit of using Schmitt-trigger as the buffer instead

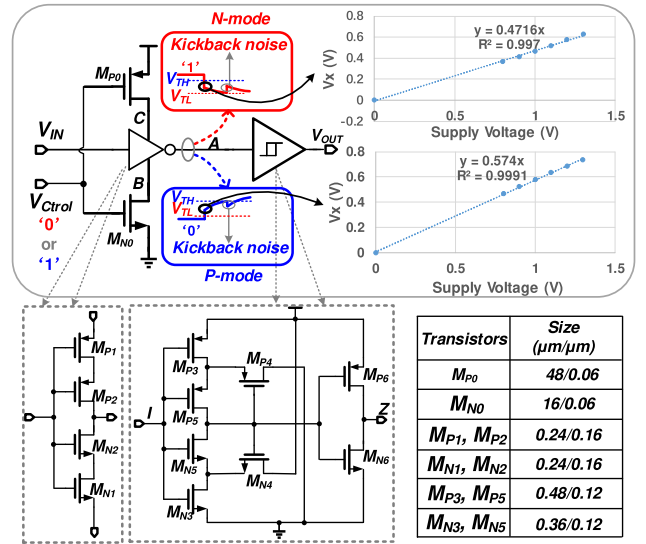


Fig. 8. Diagram of the proposed reconfigurable delay cell along with the simulated V_X versus V_{DD} plot.

of typical inverters is that the hysteresis can protect the delay cell from kickback noise that occurs when the buffer flips over in both modes.

In the proposed Schmitt-trigger-based delay cell, only regular-threshold-voltage (RVT) transistors are required, which reduces the fabrication cost. The header M_{P0} and the footer M_{N0} are designed with large size to provide a nanoampere-level leakage current at room temperature. In addition, the inverter is cascoded to ensure that the open resistance is far larger than the “disabled” header or footer, making that almost all the leakage current is used to pull down or pull up the node A. Non-minimum lengths are chosen for M_{P3} , M_{P4} , M_{N3} , and M_{N4} for better matching. Some key sizes of the transistors are listed in Fig. 8.

After fixing the sizes of the power-gated inverter and its load transistors M_{P3} , M_{P5} , M_{N3} , and M_{N5} , the balanced

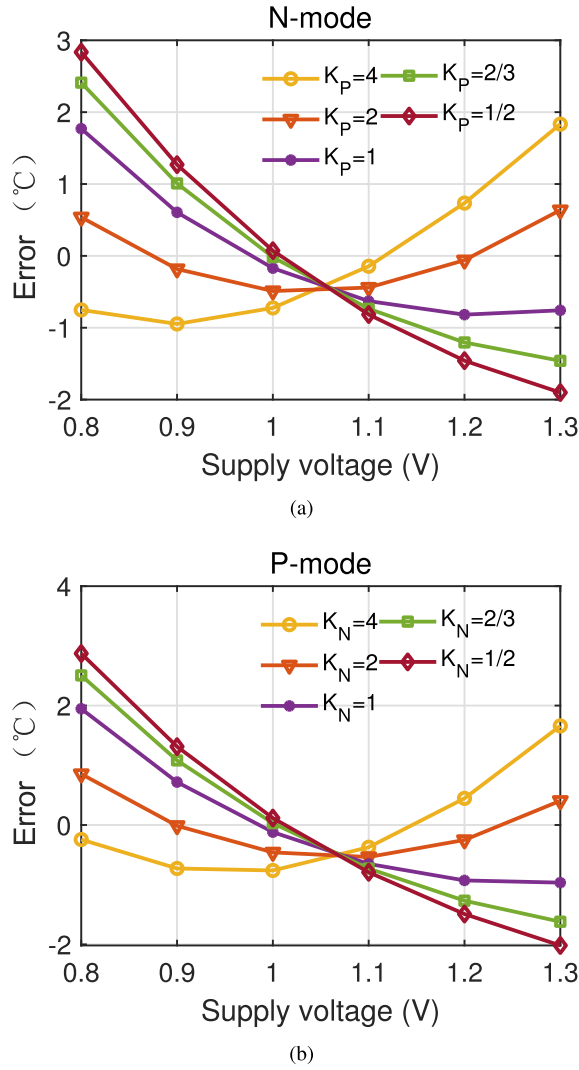


Fig. 9. (a) Simulated supply sensitivity for the N-mode sensor at different values of K_P at 25 $^{\circ}\text{C}$. (b) Simulated supply sensitivity for the P-mode sensor at different values of K_N at 25 $^{\circ}\text{C}$.

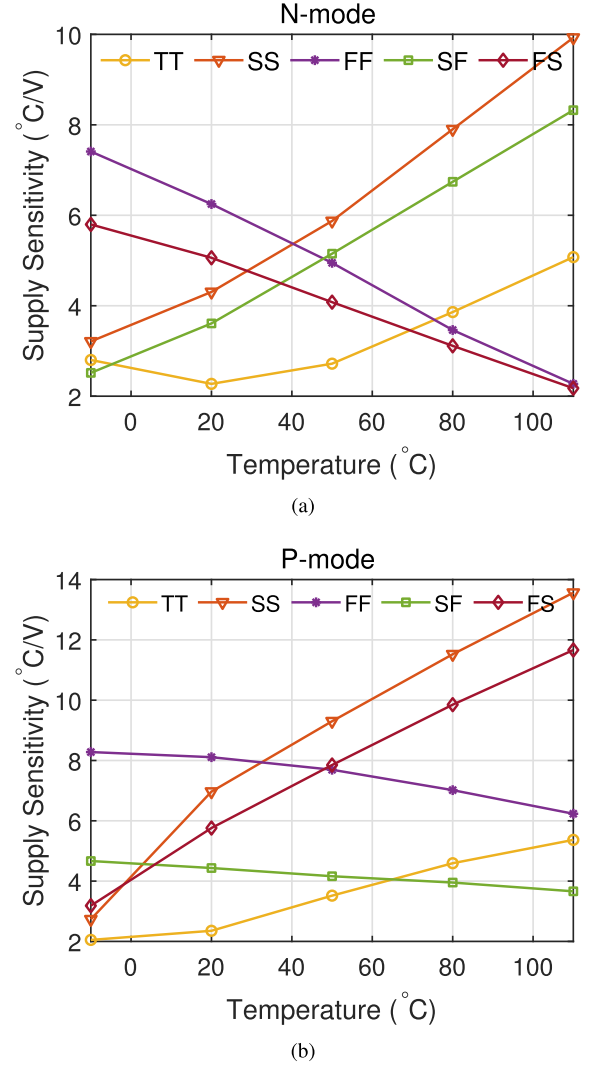


Fig. 10. (a) Simulated supply sensitivity for the N-mode sensor at different temperatures and corners. (b) Simulated supply sensitivity for the P-mode sensor at different temperatures and corners.

voltage in the quick charging phase V_X is also determined, as shown in Fig. 8 (a_1 is approximately 0.47 in the N-mode and 0.57 in the P-mode). Due to the hysteresis of the Schmitt trigger, the supply sensitivity of the delay cell in each working mode can be suppressed separately by optimizing the supply sensitivities of V_{TL} and V_{TH} . V_{TL} and V_{TH} can be estimated by [24]

$$V_{TL} \approx \frac{\sqrt{K_P}(V_{DD} - V_{TP})}{1 + \sqrt{K_P}} \quad (13)$$

$$V_{TH} \approx \frac{V_{DD} + \sqrt{K_N} \cdot V_{TN}}{1 + \sqrt{K_N}} \quad (14)$$

where K_P is the ratio of W_{P3}/L_{P3} to W_{P4}/L_{P4} , K_N is the ratio of W_{N3}/L_{N3} to W_{N4}/L_{N4} , and V_{TN} and V_{TP} are the threshold voltages of NMOS and PMOS transistors, respectively. Although (13) and (14) are the equations for long-channel transistors and the absolute value is not precise for Schmitt triggers in nanometer technologies, it shows that the supply sensitivities of V_{TL} and V_{TH} are set by K_P

and K_N independently. Fig. 9 shows the simulated supply sensitivity at 25 $^{\circ}\text{C}$ with different values of K_P and K_N for both working modes. It shows that when $K_P = 2$ and $K_N = 2$, the supply sensitivity can be reduced to 2.25 $^{\circ}\text{C}/\text{V}$ and 2.60 $^{\circ}\text{C}/\text{V}$ from 0.8 to 1.3 V in the N-mode and P-mode, respectively. The impact of process variations on the supply sensitivity under different temperatures is evaluated for both working modes. As shown in Fig. 10, the worst cases occur at SS corner and high temperatures for N and P modes, which are 10 $^{\circ}\text{C}/\text{V}$ and 13.6 $^{\circ}\text{C}/\text{V}$, respectively. Also, the impact of random mismatches on the supply sensitivity is evaluated by 1000 Monte Carlo simulations. As shown in Fig. 11, the standard deviation of the supply sensitivity for N/P mode is 0.16/0.24 $^{\circ}\text{C}/\text{V}$ at 25 $^{\circ}\text{C}$ with 500-mV dc supply voltage variations from 0.8 to 1.3 V, which is thus relatively small compared to the nominal sensitivity.

The output frequency of the reconfigurable LDRO is simulated at different corners from -10°C to 110 $^{\circ}\text{C}$, which is a regular operating range for thermal management [5]. As shown in Fig. 12, the temperature dependence of the

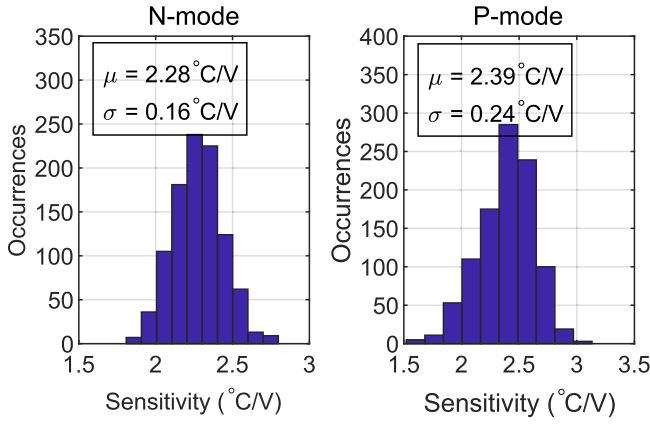


Fig. 11. Simulated supply sensitivity with 1000 Monte Carlo mismatched simulations at TT corner, 25 °C with 500-mV dc supply voltage variation from 0.8 to 1.3 V.

oscillating frequency well matches with the analysis in Section II. It shows that the residual non-linear errors can be reduced to less than ± 2 °C for both working modes at all corners after fitting using (7). As shown in Fig. 13, the effect of the random mismatch on the non-linearity is estimated with 200 mismatched simulations at the typical corner. It shows that the non-linearity errors are less than $+0.10/-0.18$ °C (3σ) and $+0.64/-0.36$ °C (3σ) in the N-mode and P-mode, respectively. However, when considering process and mismatch variations together, the frequencies may vary significantly in production, as shown in Fig. 14. Although the temperature dependence still follows (7), the large frequency range brings challenges for further frequency digitization, which requires an FDC with a wide input frequency range.

C. FDC

Fig. 15 presents the diagram of the FDC. The FDC is made up of two 16-bit counters and a finite-state machine (FSM). It digitizes the frequency of the sensing core with a given reference clock, which can be provided by a PLL or an RC oscillator in an SoC. One of the most frequently used methods in an FDC is to count the input signal and the reference clock simultaneously within a fixed period. However, the LDRO requires a few cycles to oscillate with a stable frequency after the EN signal, making the sensing signal Clk_sen asynchronous with the reference clock Clk_ref . In addition, due to the exponentially temperature-dependent frequency and the spread between die and die, Clk_sen has a wide frequency range, which ranges from several kilohertz at low temperatures to tens of megahertz at high temperatures. Because of the asynchronous property and the wide frequency range, it is difficult to count Clk_sen and Clk_ref with the same time period, causing a large quantization error at low temperatures.

To solve this issue, a faster reference clock Clk_ref is applied in this article. The timing diagram of the FDC is also shown in Fig. 15. First, Counter1 connected to Clk_sen is triggered by the start signal. Counter2 connected to Clk_ref will not start counting until the first rising edge of Clk_sen after the start signal [25]. As to the end of the conversion, the FDC outputs the LDRO cycles N_{sen} and the duration of

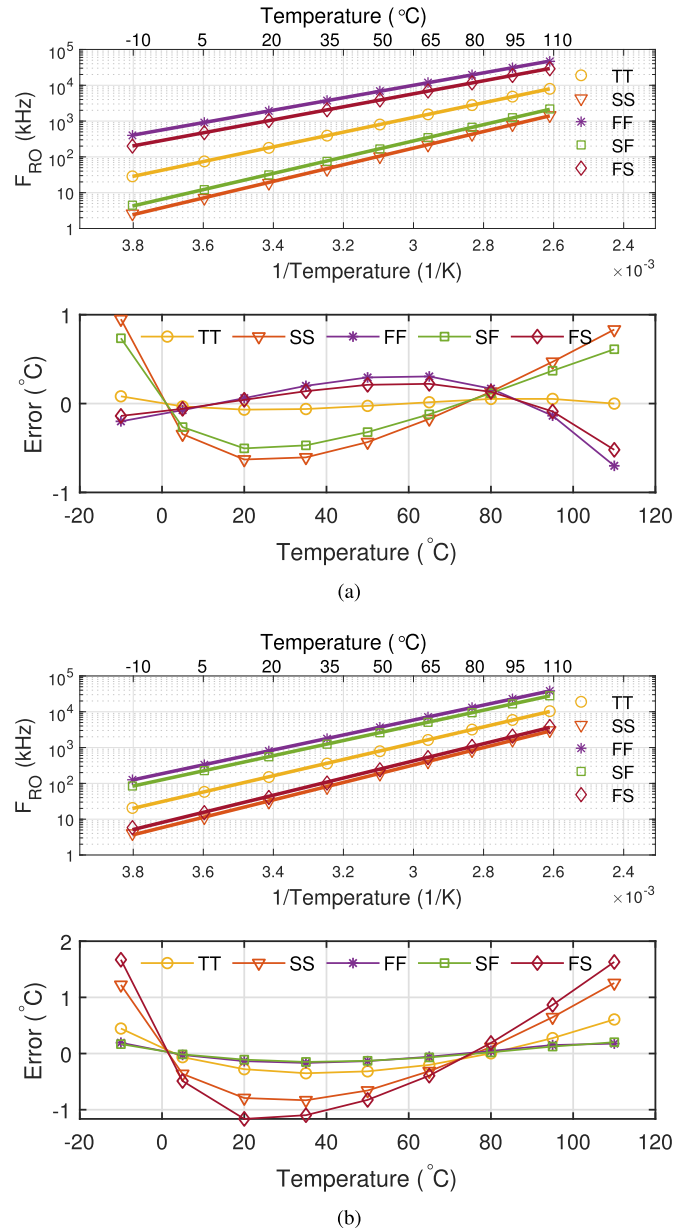


Fig. 12. (a) Simulated frequency for the N-mode sensor at different corners (top) and the corresponding non-linear error after fitting using (7) at different corners (bottom). (b) Simulated frequency for the P-mode sensor at different corners (top) and the corresponding non-linear error after fitting using (7) at different corners (bottom).

those cycles N_{ref} within a fixed number of reference cycles N_{max} [14], and thus, the quantization error can be reduced to less than ± 1 Clk_ref cycle. The sensing frequency is finally computed as $N_{\text{sen}}/N_{\text{ref}} \cdot F_{\text{ref}}$, where F_{ref} is the frequency of Clk_ref . If a 50-MHz reference clock is applied, the frequency range of the FDC is from 0.763 kHz to 50 MHz. To make this FDC fixable, a programmable reference clock can be applied when integrating this sensor into an SoC for thermal management.

D. Other Error Sources

Since both the LDRO and the FDC serve as low-pass filters, the jitter caused by high-frequency noise from the transistors

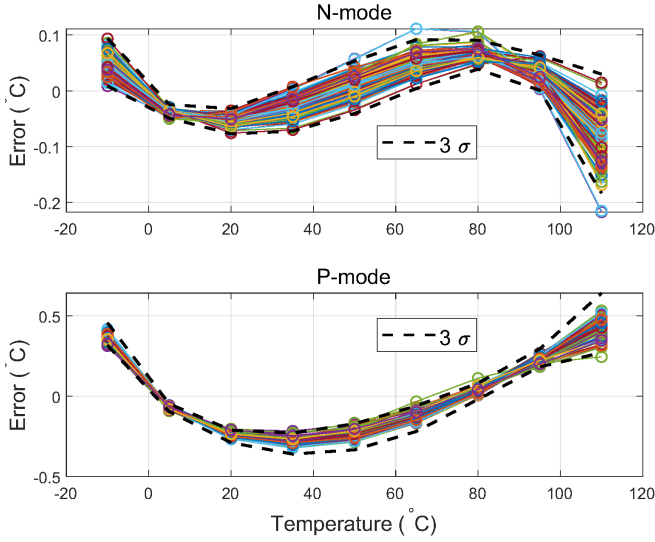


Fig. 13. Simulated nonlinearity with 200 Monte Carlo mismatched simulations at TT corner.

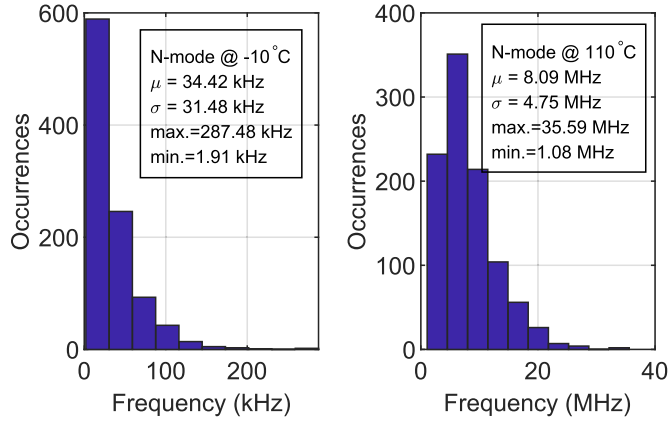


Fig. 14. Simulated frequencies with 1000 Monte Carlo simulations when considering both process and mismatch variations.

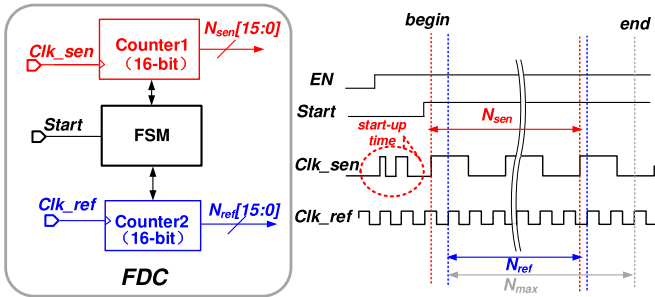


Fig. 15. Block and timing diagram of the FDC.

and supply voltage can be averaged out. Furthermore, thanks to the exponential temperature dependence of the LDRO, errors caused by supply variations, jitter in the LDRO and uncertainty of the reference clock, are further attenuated by the logarithmic frequency-to-temperature conversion [15]. For example, an RC oscillator-based reference clock with a temperature coefficient of 100 ppm/°C will only cause a relative

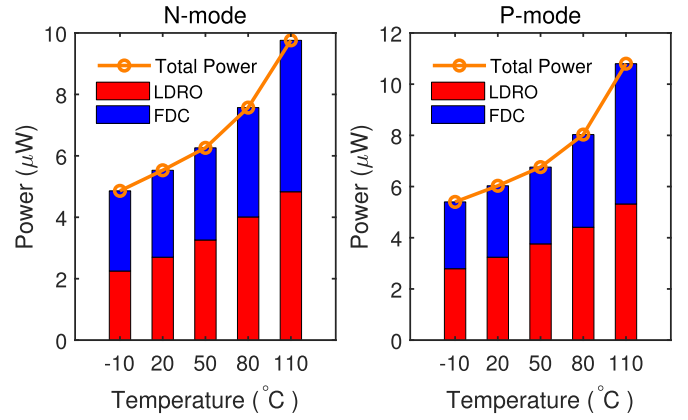


Fig. 16. Simulated power consumption of the sensor in N-/P-mode at TT corner.

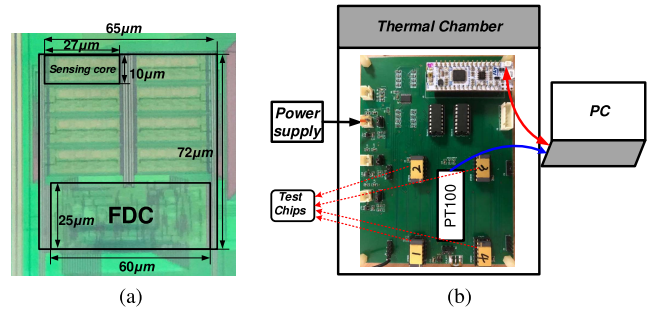


Fig. 17. (a) Die micrograph of the proposed temperature sensors. (b) Test setup for the proposed temperature sensors.

inaccuracy of 0.2%, assuming that the temperature sensitivity of the LDRO is 5%/°C.

E. Power Consumption

Operating under a 0.9-V supply voltage and a 50-MHz reference clock, the sensor working in the N-mode consumes a total power consumption of 5.5 μ W at room temperature according to simulations. The power consumed by the LDRO is approximately 2.7 μ W and the FDC dissipates 2.8 μ W. Similarly, it dissipates 6.1 μ W when working in the P-mode at room temperature. The power consumption of the LDRO is dominated by Schmitt triggers, in which the input devices operate in the saturated region during the charging or discharging phase. As shown in Fig. 16, the power rises at high temperatures due to the increased leakage current and the sensing frequency.

IV. MEASUREMENT RESULTS

The proposed temperature sensor is fabricated in a standard 55-nm CMOS digital process, as shown in Fig. 17(a). The chip contains eight sensing cores (each consisting of a reconfigurable LDRO), a multiplexer, and a shared FDC. A single sensor, including the multiplexer and FDC, occupies 1770 μ m², which is mainly dominated by the all-digital FDC. Each LDRO functioning as the sensing core occupies a die area of only 270 μ m². Thanks to the all-digital architecture, the area can be scaled down easily with CMOS processes.

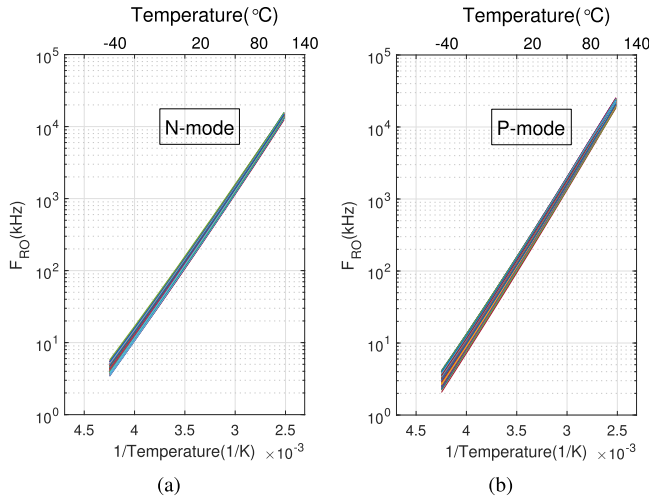


Fig. 18. (a) Measured frequencies versus temperature of 64 sensing cores in N-mode. (b) Measured frequencies versus temperature of 64 sensing cores in P-mode.

For the temperature sensing characterization, eight chips (containing 64 sensing cores in total) from one batch are packaged in ceramic DIP packages and measured with a 0.9-V supply voltage and a 50-MHz reference clock over the temperature range from $-40\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$. As shown in Fig. 17(b), the sensors are placed next to a reference sensor (platinum Pt-100 resistor) for reference. An onboard MCU (STM32L011K4) is used for data access. They are put inside a metal box that serves as a low-pass thermal filter and measured in the thermal chamber.

A. Accuracy and Calibration

Fig. 18 shows the measured frequencies in two working modes: 4 kHz–15 MHz for N-mode sensors (around $5.6\text{ }^{\circ}\text{C}$ at $20\text{ }^{\circ}\text{C}$) and 3.5 kHz–23 MHz for P-mode sensors (around $6.1\text{ }^{\circ}\text{C}$ at $20\text{ }^{\circ}\text{C}$). The absolute frequencies vary among different chips. However, they all show good linearity versus temperature after linearization according to (7). The two coefficients a and b in (7) can be calculated by two-point calibration for each sample. The measured accuracy after two-point calibration with/without systematic error removal is plotted in Fig. 19. It shows that the sensors have measured 3σ inaccuracies of less than $+1.78/-1.66\text{ }^{\circ}\text{C}$ in the N-mode and $+1.78/-1.37\text{ }^{\circ}\text{C}$ in the P-mode from $-40\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$ after two-point calibration. To further improve the accuracy, fixed third-order polynomial fitting can be applied to remove the systematic error [5], [9]. The errors are reduced to $\pm 0.70\text{ }^{\circ}\text{C}$ (3σ , N-mode) and $\pm 0.94\text{ }^{\circ}\text{C}$ (3σ , P-mode) from $-40\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$ after the systematic error removal. For the target temperature range from $-10\text{ }^{\circ}\text{C}$ to $110\text{ }^{\circ}\text{C}$ for the thermal management application, the proposed sensors show a measured inaccuracy of $\pm 0.54\text{ }^{\circ}\text{C}/\pm 0.78\text{ }^{\circ}\text{C}$ (3σ , N/P-mode). Fig. 20 shows the results with one-point calibration and systematic error removal for both working modes. Even after low-cost one-point calibration, an inaccuracy of $\pm 1.38\text{ }^{\circ}\text{C}/\pm 1.64\text{ }^{\circ}\text{C}$ (3σ , N/P-mode) from $-10\text{ }^{\circ}\text{C}$ to $110\text{ }^{\circ}\text{C}$ is also sufficient for the thermal management application.

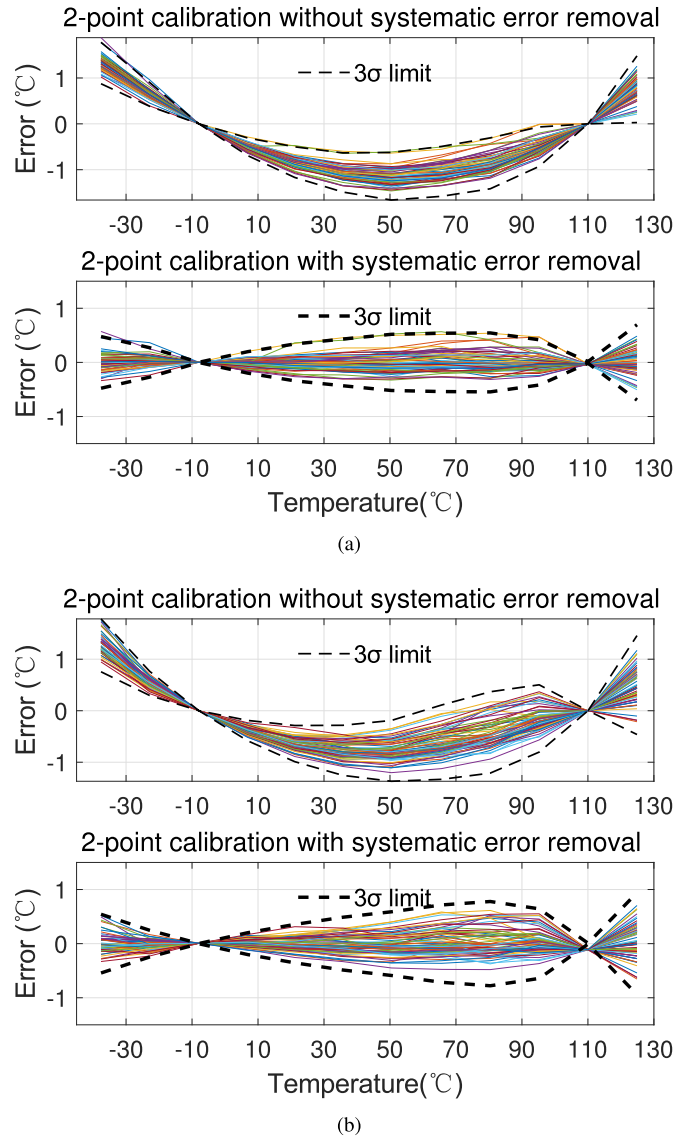


Fig. 19. (a) Measured inaccuracy of the N-mode sensor after two-point calibration with/without systematic error removal. (b) Measured inaccuracy of the P-mode sensor after two-point calibration with/without systematic error removal.

B. Supply Sensitivity

Thirty-two sensors are measured under a supply voltage from 0.8 to 1.3 V, which is mainly limited by the all-digital synthesized FDC and the data interface (SPI) for the testing. As shown in Fig. 21, the sensors have a supply sensitivity of $2.53\text{--}5.22\text{ }^{\circ}\text{C}/\text{V}$ (Min-Max) and $2.84\text{--}5.76\text{ }^{\circ}\text{C}/\text{V}$ (Min-Max) at $20\text{ }^{\circ}\text{C}$ for N-mode and P-mode, respectively. Due to the process variations, the optimal supply sensitivity suppression range shifts a little for the different working modes. If a smaller supply range is considered for the corresponding working mode separately, the measured supply sensitivity is $1.71\text{--}4.37\text{ }^{\circ}\text{C}/\text{V}$ from 0.9 to 1.3 V (N-mode) and $1.63\text{--}4.38\text{ }^{\circ}\text{C}/\text{V}$ from 0.8 to 1.2 V (P-mode). The measured supply sensitivities at different temperatures are also summarized in Table I. It shows that the averaged supply sensitivity shifts with temperature but stays within an acceptable range ($\leq 12.7\text{ }^{\circ}\text{C}/\text{V}$) for each mode. Table I also reveals that the

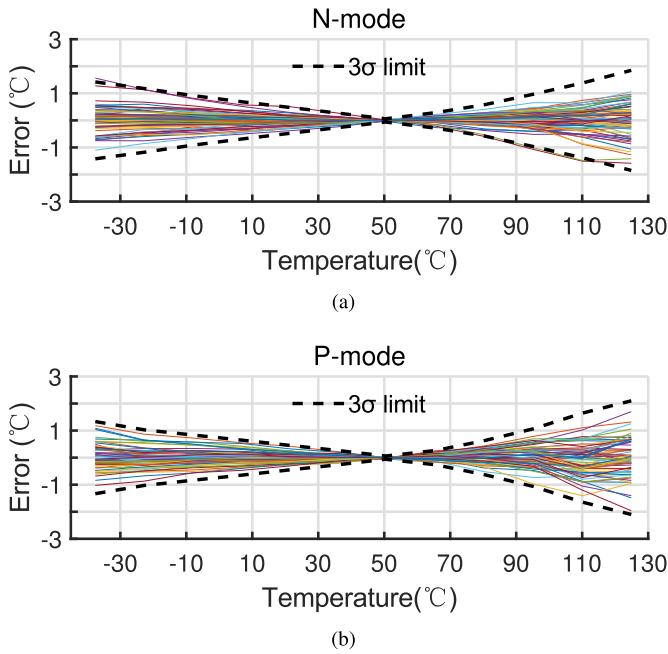


Fig. 20. (a) Measured inaccuracy of the N-mode sensor after one-point calibration and systematic error removal. (b) Measured inaccuracy of the P-mode sensor after one-point calibration and systematic error removal.

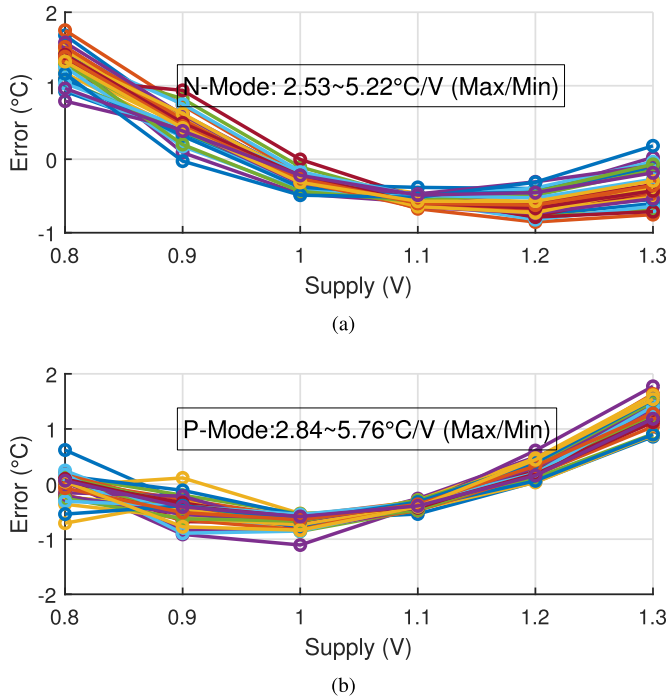


Fig. 21. (a) Measured supply sensitivity of the N-mode sensor from 0.8 to 1.3V at 20 °C. (b) Measured supply sensitivity of the P-mode sensor from 0.8 to 1.3V at 20 °C.

P-mode sensor has better supply rejection at high temperatures, while the N-mode sensor is better at low temperatures. If the sensor is configured into P-mode at high temperatures and into N-mode at low temperatures, the supply sensitivity can be further reduced to less than 5.58 °C/V over the full temperature range.

TABLE I
MEASURED SUPPLY SENSITIVITY AT DIFFERENT TEMPERATURES
FOR BOTH N-/P-MODE SENSORS

Type		20 °C	-40°C	50°C	125°C*
N	p-p(°C/V)	2.53-5.22	1.12-3.94	4.11-7.16	10.26-15.35
	Avg.(°C/V)	3.87	1.83	5.70	12.70
P	p-p(°C/V)	2.84-5.76	3.94-8.16	2.22-4.73	0.8-5.58
	Avg.(°C/V)	4.07	5.07	3.40	2.60

* The value at 125°C were measured from 0.8 V to 1.2 V due to the limitation of the SPI. The rest were measured from 0.8 V to 1.3 V.

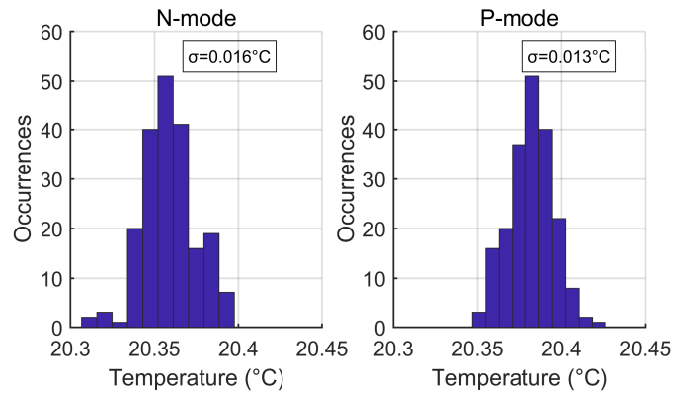


Fig. 22. Measured histogram of 200 sensor readings in both modes.

C. Resolution and FoM

Fig. 22 shows a histogram of 200 measurements at room temperature, and the sensor achieves a thermal-noise-limited resolution of 16/13 mK in N/P modes with a conversion time of 1.31 ms. Since the sensors are integrated with other digital circuits and share the same power supply in a larger chip, the power consumption is estimated by multiplying the measured total power by the simulated power ratio of this sensor to the total chip. Measured with a 0.9-V supply voltage at room temperature, the power consumption is 9.3 and 9.8 μ W in the N-mode and P-mode, respectively. Considering the resolution, the conversion time, and the power consumption, the sensor shows a resolution FoM of 3.1 $\text{pJ}\cdot\text{K}^2/2.2 \text{ pJ}\cdot\text{K}^2$ (N-/P-mode).

D. Comparison of N/P Mode and Future work

As can be seen from both the simulated and measured results, the N-mode sensor has better linearity and accuracy when compared to the P-mode sensor. This can be explained by the fact that an NMOS transistor has a larger unit subthreshold current due to its greater mobility, which mitigates the impact of other kinds of leakage, such as p-n junction leakage and gate leakage in the “disabled” transistor. In addition, due to the greater mobility and leakage current density, the “disabled” transistor in the N-mode sensor is smaller than that in the P-mode sensor, which means that it can be more area efficient when only the N-mode sensor is applied. On the other hand, the P-mode sensor shows a slightly better rms resolution

TABLE II
PERFORMANCE SUMMARY AND COMPARISON WITH THE STATE-OF-THE-ART DESIGNS

Parameters	This work		JSSC'16 [5]	SSC-L'18 [14]	TCAS-I'13 [13]	ISSCC'17 [15]	ESSCIRC'18 [3]	JSSC'17 [12]	ISSCC'18 [10]
	N-mode	P-mode							
Type	MOS		MOS	MOS	MOS	MOS	BJT	TD	RES
Technology	55 nm		65 nm	28 nm	65 nm	180 nm	65 nm	40 nm	65 nm
Area (μm^2)	270 ^a , 1770 ^b		4000 ^b	225 ^a , 1044 ^c	8000 ^b	8865 ^b	3000 ^b	1650 ^b	7000 ^a
External references	Clock ref.		None	LDO Iref., Clock ref.	None	Clock ref.	None	None	Clock ref.
Supply (V)	0.8-1.3	0.8-1.3	0.85-1.05	0.62-1.2	1.2	1.2	1.2-1.8	0.9-1.2	0.85-1.05
Temp.Range ($^{\circ}\text{C}$)	-40-125	-40-125	0-100	-5-85	0-110	-20-100	-10-110	-40-125	-40-85
Inaccuracy ($^{\circ}\text{C}$)	± 0.70 3 σ	± 0.94 3 σ	± 0.9 Max-Min	-1.4/1.3 Max-Min	± 1.5 Max-Min	-0.22/0.19 Max-Min	± 1.35 3 σ	± 0.75 3 σ	± 0.35 3 σ
Calibration points	2		2	1	1	2	2	1	2
Resolution (mK)	16	13	300	760	940	73	130	360	2.8
Conversion time (ms)	1.31	1.31	0.022	0.036	0.002	8	4.1	1	1
Power (μW)	9.3 ^{b,d}	9.8 ^{b,d}	154 ^b	56 ^c	500	0.075 ^b	111.8	2500	68 ^a
Energy (nJ/conv.)	12.2 ^{b,d}	12.8 ^{b,d}	3.39 ^b	2.0 ^c	1.0	0.6 ^b	458.4	2500	68 ^a
Supply sensitivity ($^{\circ}\text{C}/\text{V}$)	2.53-5.22 ^f	2.84-5.76 ^f	34	1.9 ^{e,f}	N/A ^e	0.13 ^{e,f}	3-14	2.8	0.5 ^f
Res.FoM (pJ·K ²)	3.1	2.2	305	1164	942	3.2	7700	3.2e5	0.53

^a Sensing core only.

^b Sensing core+ADC/FDC, excluding further processing logic, such as logarithmic operation, decimation, or non-linearity correction. The off-chip logarithmic operation and non-linearity correction in this work is approximately 0.037 mm² and can be shared for all sensors or be done by firmware.

^c Normalized per probe, including the current reference and processing logic.

^d Multiplying the measured total power by the simulated power ratio of this sensor to the total chip.

^e Supply regulators or native IO devices are used.

^f Supply sensitivity around room temperature.

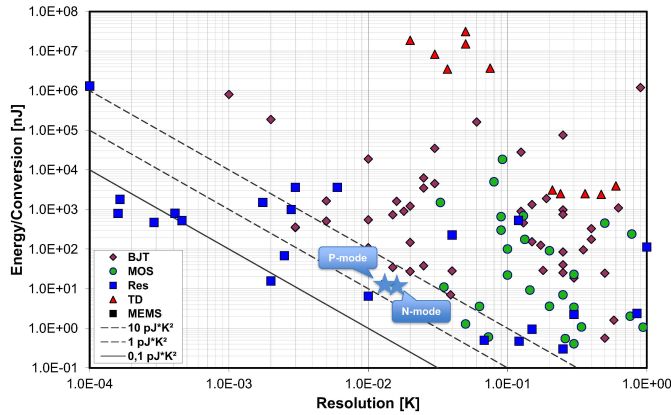


Fig. 23. Graphical comparison of the proposed sensor in the energy-per-measurement versus resolution plot [6].

than the N-mode one because of the stand-alone n-well as well as the larger temperature sensitivity. For the application of thermal management, where the resolution is not critical, the N-mode sensor is a better choice.

In the future work, the supply sensitivity can be further improved if the threshold voltages V_{TL} and V_{TH} are designed to be adjustable to compensate process variations. Also, the resolution of the FDC can be degraded, making this sensor to be more in line with the needs of on-chip thermal monitoring.

E. Comparison to Previous Work

Table II summarizes this article and compares it with state-of-the-art designs. Thanks to the concise digital

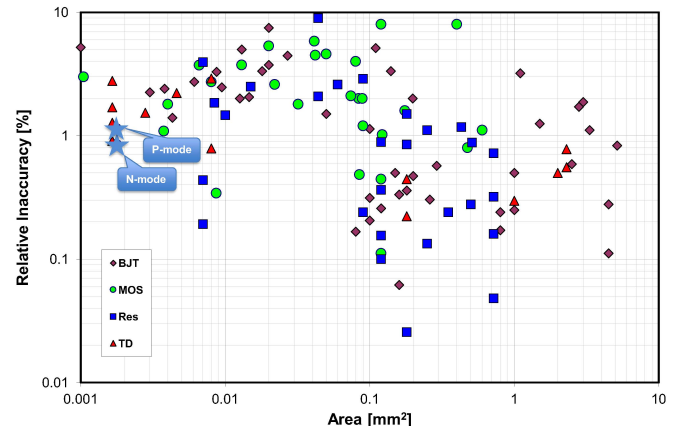


Fig. 24. Comparison of the proposed sensor in the relative inaccuracy versus area plot [6].

architecture, the proposed sensor occupies a small silicon area of 1770 μm^2 . It shows a low supply sensitivity without any voltage regulator or native IO devices. In addition, it also achieves better energy efficiency compared to the compact BJT- and TD-based designs in nanometer CMOS with similar or better accuracy [3], [12]. Graphical comparisons of the resolution FoM, the relative accuracy, and the area with the previously published temperature sensors [6] are shown in Figs. 23 and 24.

V. CONCLUSION

A leakage-based digital CMOS temperature sensor with reduced supply sensitivity is analyzed and designed in a standard 55-nm CMOS process. The prototype sensor occupies

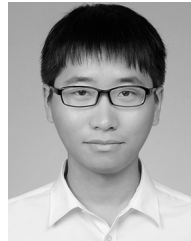
a silicon area of $1770 \mu\text{m}^2$. It can operate at a supply voltage from 0.8 to 1.3 V with the supply sensitivities of $2.53\text{--}5.22 \text{ }^\circ\text{C/V}$ and $2.84\text{--}5.76 \text{ }^\circ\text{C/V}$ in two working modes at room temperature, respectively. Measurement results show that it achieves an inaccuracy of $\pm 0.70 \text{ }^\circ\text{C}/\pm 0.94 \text{ }^\circ\text{C}$ (3σ , N/P-mode) from $-40 \text{ }^\circ\text{C}$ to $125 \text{ }^\circ\text{C}$ with two-point calibration. Due to their high accuracy, small area, high energy efficiency, and reduced supply sensitivity, the sensors are suitable for thermal management applications in nanometer CMOS technologies.

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circuits' design and temperature sensor design.

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